

Name _____

Date March 5, 2025

Total
28
/30

1. Differentiate the following functions. Express answers in radical form.

$$\begin{aligned} \text{b) } f(x) &= \sqrt[3]{64x^4} \\ &= (64x^4)^{\frac{1}{3}} \\ f'(x) &= \left(\frac{1}{3}\right)(256x^3)(64x^4)^{-\frac{2}{3}} \\ &= \frac{256x}{3\sqrt[3]{(64x^4)^2}} \quad \frac{16}{3\sqrt[3]{x}} \end{aligned}$$

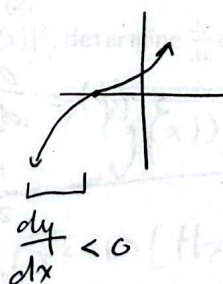
$$\begin{aligned} \text{b) } y &= \frac{2-x}{x^2+1} \frac{f(x)}{g(x)} \\ f'(x) &= \frac{(-1)(x^2+1) - (2x)(2-x)}{(x^2+1)^2} \quad 2.5 \\ &= \frac{-x^2 - 1 - 4x + 2x^2}{(x^2+1)^2} \\ &= \frac{x^2 - 4x - 1}{(x^2+1)^2} \end{aligned}$$

$$\begin{aligned} \text{d) } g(x) &= x\sqrt{x-1} \\ &= x(x-1)^{\frac{1}{2}} \\ g'(x) &= (1)(x-1)^{\frac{1}{2}} + \left(\frac{1}{2}\right)(1)(x-1)^{-\frac{1}{2}}(x) \\ &= \frac{1}{\sqrt{x-1}} + \frac{x}{2\sqrt{x-1}} \\ &= \frac{2+x}{2\sqrt{x-1}} \end{aligned}$$

$$\begin{aligned} \text{d) } y &= (x^2 + 4x + 4)^2 \\ y' &= 2(2x+4)(x^2+4x+4) \quad 3.5 \\ &= 2(2x^3 + 8x^2 + 8x + 4x^2 + 16x + 16) \\ &= 2(2x^3 + 12x^2 + 24x + 16) \\ &= 4x^3 + 24x^2 + 48x + 32 \end{aligned}$$

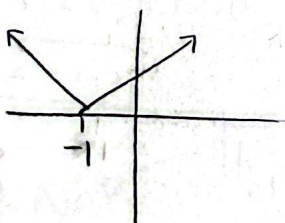
2. Determine the intervals of $y = (x^2 + 4x + 4)^2$ where $\frac{dy}{dx} < 0$. You may use your answer from question 1d)

$$\begin{aligned} y' &= 2(2x+4)(x^2+4x+4) \\ &= 2(x+2)(x+2)^2 \\ &= 2(x+2)^3 \leftarrow x = -2 \\ \text{order 3 } CC = (+) \end{aligned}$$



$$\frac{dy}{dx} < 0 \text{ when } (-\infty, -2) \quad (2)$$

3. State a function $f(x)$ that is continuous on $x \in [-\infty, \infty]$ but is not differentiable at $x = -1$. Demonstrate using a diagram.



$$\begin{aligned} f(x) &= |x+1| \\ \text{non-diff @ } x &= -1 \\ \text{due to corner but} \\ \text{cont. } x \in \mathbb{R} \end{aligned}$$

(1)

9

4. Determine the derivative of $f(x) = [(3x^2 + 1)^3 + 1]^2$. Express your answer in a simplified factored form.

$$\begin{aligned} f(x) &= [(3x^2 + 1)^3 + 1]^2 \quad f'(x) = 2[(3x^2 + 1)^3 + 1] \cdot 3(6x)(3x^2 + 1)^2 \\ &= 2[(3x^2 + 1)^3 + 1] \cdot 18x(3x^2 + 1)^2 \\ &= 2[(3x^2 + 1)^3 + 1] \cdot 18x(3x^2 + 1)^2 \\ &= 36x[(3x^2 + 1)^3 + 1](3x^2 + 1)^2 \end{aligned}$$

(2)

5. If $y = 2u^3 + 3u^2$, and $u = \sqrt{2x-1}$, use the chain rule to find $\frac{dy}{dx}$ at $x = 5$.

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} & \frac{dy}{dx} &= \left(\frac{1}{\sqrt{2x-1}} \right) (6u^2 + 6u) \\ y &= 2u^3 + 3u^2 & u &= \sqrt{2x-1} \\ y' &= 6u^2 + 6u & &= (2x-1)^{-\frac{1}{2}} \\ @ x=5, u=3 & & u' &= \frac{1}{2}(2)(2x-1)^{-\frac{1}{2}} \\ & & &= \frac{1}{\sqrt{2x-1}} \\ & & &= \frac{1}{\sqrt{2(5)-1}} \\ & & &= \frac{1}{\sqrt{9}} = \frac{1}{3} \\ & & &= \frac{1}{3} \cdot (3) \cdot (72) \\ & & &= 24 \end{aligned}$$

6. The function $s(t) = (t^3 + t^2)^{\frac{1}{2}}, t \geq 0$, represents the displacement s , in metres, of a particle moving along a straight line over t seconds. Determine $s(4)$ and $s'(4)$ and interpret both values for this situation. Express your answers to the nearest tenth.

$$\begin{aligned} s(t) &= (t^3 + t^2)^{\frac{1}{2}} & s'(t) &= \frac{dy}{dx} (t^3 + t^2)^{\frac{1}{2}} \\ s(4) &= (4^3 + 4^2)^{\frac{1}{2}} & &= \frac{1}{2} (3t^2 + 2t) (t^3 + t^2)^{-\frac{1}{2}} \\ &= 4\sqrt{5} & &= \frac{3t^2 + 2t}{2\sqrt{t^3 + t^2}} \\ &= 8.9 \text{ m} \end{aligned}$$

$s'(4) = 3.1 \text{ m/s}$ (reps the velocity @ 4 secs) and
 $s(4) = 8.9 \text{ m}$ (reps. the displacement @ 4 secs)

$$\begin{aligned} s'(4) &= \frac{3(4)^2 + 2(4)}{2\sqrt{(4)^3 + (4)^2}} \\ &= \frac{52}{2\sqrt{64 + 16}} \\ &= \frac{52}{2\sqrt{80}} \rightarrow 3.1 \text{ m/s} \end{aligned}$$

8.5

perp. to tangent slope

7. The normal to the graph of $f(x) = \frac{x^2-1}{(x+2)}$ at the point (a,b) is represented by $2y - x - 1 = 0$. Determine the values of a and b .

$$2y - x - 1 = 0$$

$$y = \frac{1}{2}x + \frac{1}{2}$$

$$\text{tan. slope} = -2$$

$$f(x) = \frac{x^2-1}{(x+2)}$$

$$f'(x) = \frac{(2x)(x+2) - (1)(x^2-1)}{(x+2)^2}$$

$$= \frac{2x^2 + 4x - x^2 + 1}{(x+2)^2}$$

$$a = -1 \quad a = -3$$

$$f(x) = 0$$

$$f(x) = -8$$

$$-2 = \frac{x^2 + 4x + 1}{(x+2)^2}$$

$$-2(x^2 + 4x + 4) = x^2 + 4x + 1$$

$$\text{plug in } a = x$$

$$-2(a^2 + 4a + 4) = a^2 + 4a + 1$$

$$-2a^2 - 8a - 8 = a^2 + 4a + 1$$

$$-3a^2 - 12a - 9 = 0$$

$$a = \frac{-11 \pm 3}{2}$$

$$\text{if } a = -1, b = 0$$

$$\text{if } a = 3, b = -8$$

point is not on the graph

8. If $f(-2) = -4$, and $f'(-2) = 8$, and $y = (x^2 - 1)[f(x)]^2$, determine $\frac{dy}{dx}$ at $x = -2$

$$f(-2) = -4 \quad f'(-2) = 8$$

$$y = (x^2 - 1)(f(x))^2$$

$$\textcircled{1} f'(x) = 2x$$

$$\textcircled{2} f(-2) = -4 \quad f'(-2) = 8$$

$$y' = 2(-4) \cdot 8$$

$$\boxed{= -64}$$

IGNORE !!

$$h'(x) = f'(g(x)) \cdot g'(x)$$

$$y = (x^2 - 1)[f(x)]^2 \quad [f(x)]^2$$

$$y' = (2x)(f(x))^2 + 2(f'(x))(x^2 - 1)$$

$$\textcircled{a} x = -2$$

$$y' = 2(-2)(-4)^2 + 2(8)(-4)((-2)^2 - 1)$$

$$= -64 + (-192)$$

$$\boxed{= -256}$$

5.1

9. Examine the differentiability of $f(x) = \frac{x-2}{x^2+4}$.

① $f(x)$ is cont. $\forall x \in \mathbb{R}$

② $f'(x) = \frac{(1)(x^2+4) - (2x)(x-2)}{(x^2+4)^2}$

$$= \frac{x^2+4-2x^2+4x}{(x^2+4)^2}$$

$$= \frac{-x^2+4x+4}{(x^2+4)^2}$$

$\forall x \in \mathbb{R}$ for $f'(x)$

$\therefore f(x)$ is cont. and differentiable over $\mathbb{R}$

(2)

10. Determine the value of m and n such that $f(x) = \begin{cases} 3-x, & x \leq 1 \\ ax^2+bx, & x > 1 \end{cases}$ is differentiable.

① must be cont.

② must have derivative

① $3-x, x \leq 1$

① $f'(x) = -1$

② $ax^2+bx, x > 1$

② $f'(x) = 2ax+b$

$3-x = ax^2+bx$

plug in $x=1$

$-1 = 2ax+b$

plug in $x=1$

$3-(1) = a(1)^2+b(1)$

$-1 = 2a(1)+b$

$2 = a+b$

$-1 = 2a+b$

$b = 2-a$

$-1 = 2a+b$

$b = 2-a$

$-1 = 2a+(2-a)$

$b = 2-(-3)$

$-1 = 2a+2-a$

$= 5$

$-3 = a$

$a = -3$
$b = 5$

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